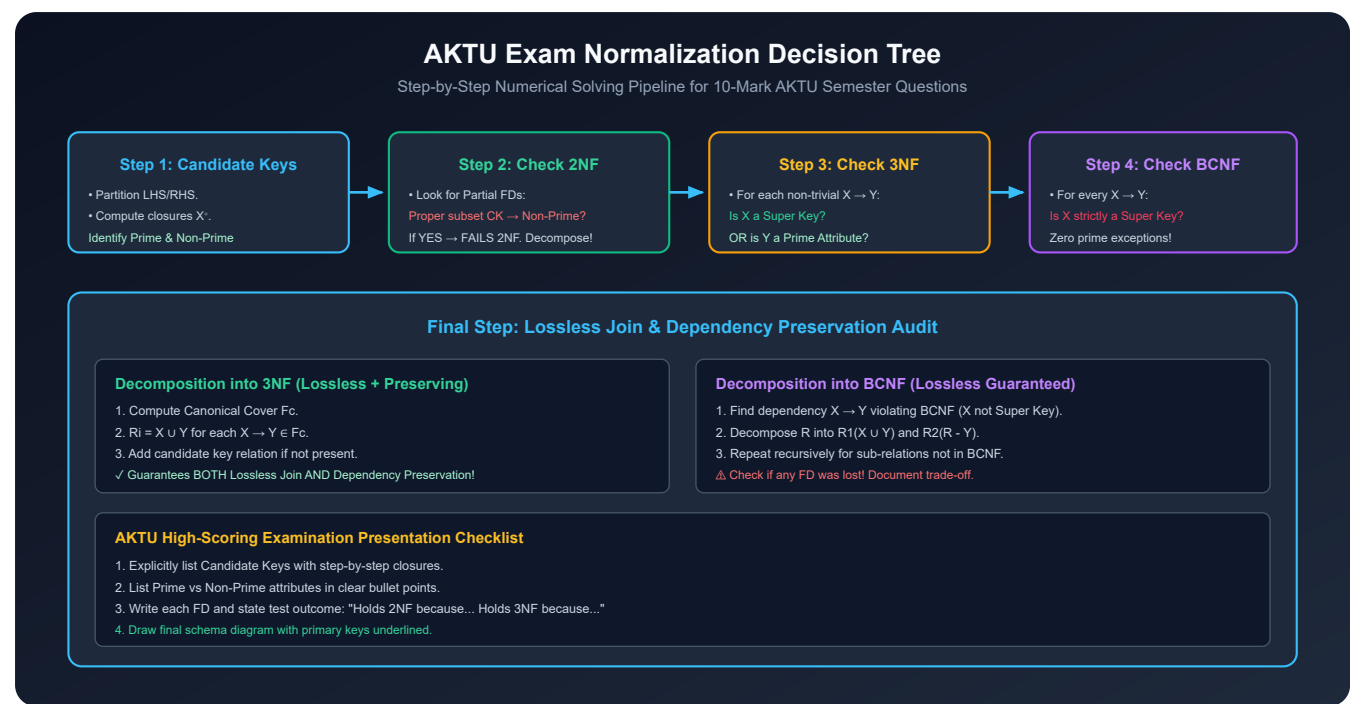


Module 12: AKTU PYQs & Solved Decompositions



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Previous Years Exam Questions (2019-2024), 7-Mark & 10-Mark Numerical Solutions, Finding Candidate Keys, Determining Highest Normal Form, Lossless 3NF/BCNF Decomposition.

1. AKTU PYQ 1 (10 Marks — 2022-23 End-Sem)

Question:

Given a relation $R(A, B, C, D, E, F)$ with the following set of functional dependencies:

$$F = \{ A$$

$$\rightarrow B, \quad C$$

$$\rightarrow DE, \quad AC$$

$$\rightarrow F \}$$

- Find all Candidate Keys of R .
- Identify Prime and Non-Prime attributes.
- Determine the highest normal form of R .
- Decompose the relation into 3NF ensuring Lossless Join and Dependency Preservation.

Solution:

Step 1: Candidate Keys

Attributes on RHS: $\{B, D, E, F\}$.

Attributes **NEVER on RHS:** $\{A, C\}$.

Essential attribute set = $\{AC\}$.

Compute closure of $\{AC\}$:

- $\{AC\}^+ = \{A, C\}$

- Using A

$B \rightarrow A \implies \{A, C, B\}$

- Using C

$DE \rightarrow AC \implies \{A, C, B, D, E\}$

- Using AC

$F \rightarrow AC \implies \{A, B, C, D, E, F\} = R$.

Since $\{AC\}^+ = R$ and no proper subset of $\{AC\}$ can determine R , **$\{AC\}$ is the ONLY Candidate Key.**

Step 2: Attribute Classification

Prime Attributes: $\{A, C\}$

Non-Prime Attributes: $\{B, D, E, F\}$

Step 3: Highest Normal Form Test

Let us test each FD in F :

1. A

$B \rightarrow A$:

- Determinant A is a proper subset of Candidate Key $\{AC\}$.

- Dependent B is a **Non-Prime Attribute**.

- This is a **Partial Dependency**!

- **Verdict:** Violates 2NF!

Highest Normal Form = 1NF.

Step 4: Decomposition into 3NF (Using Bernstein Synthesis)

1. Find Canonical Cover F_c :

- $F = \{A$

$\rightarrow B, C$

$\rightarrow D, C$

$\rightarrow E, AC$

$\rightarrow F\}$.

- Check extraneous attributes in AC

$\rightarrow F$:

- $A^+ = \{A, B\}$ (No F).
- $C^+ = \{C, D, E\}$ (No F).
- Neither is extraneous.
- Check redundant FDs: None.
- $F_c = \{A$

$\rightarrow B, \text{quad } C$

$\rightarrow D, \text{quad } C$

$\rightarrow E, \text{quad } AC$

$\rightarrow F \}$.

2. Form Relations from F_c :

- $R_1(\mathbf{A}, B)$ with Key = A (from A
 $\rightarrow B$).
- $R_2(\mathbf{C}, D, E)$ with Key = C (from C
 $\rightarrow D, C$
 $\rightarrow E$).
- $R_3(\mathbf{A, C}, F)$ with Key = $\{AC\}$ (from AC
 $\rightarrow F$).

3. Candidate Key Check:

- R_3 already contains the candidate key $\{AC\}$!

4. Final 3NF Relations:

- $\mathbf{R_1(A, B)}$, $\mathbf{R_2(C, D, E)}$, $\mathbf{R_3(A, C, F)}$.
- **Properties:** Lossless Join (✓) and Dependency Preservation (✓).

2. AKTU PYQ 2 (7 Marks — 2021-22 End-Sem)

Question:

Relation $R(A, B, C, D)$ has $F = \{AB$

$\rightarrow C, \text{quad } C$

$\rightarrow D, \text{quad } D$

$\rightarrow A \}$.

Is this relation in 3NF? Is it in BCNF? Justify your answer.

Solution:

1. Candidate Keys:

- $(AB)^+ = \{A, B, C, D\} = R \implies \mathbf{AB}$ is CK.
- $(BC)^+ = \{B, C, D, A\} = R \implies \mathbf{BC}$ is CK.
- $(BD)^+ = \{B, D, A, C\} = R \implies \mathbf{BD}$ is CK.
- Prime Attributes = $\{A, B, C, D\}$. Non-Prime = $\emptyset$.

2. **Test 3NF Condition (For every $X \twoheadrightarrow Y$:

X is Super Key OR Y is Prime):**

- AB

C : AB is Candidate Key (Super Key). $\checkmark$

- C

D : C is not Super Key, but D is **Prime Attribute** (in BD). $\checkmark$

- D

A : D is not Super Key, but A is **Prime Attribute** (in AB). $\checkmark$

- **Relation IS in 3NF!**

3. **Test BCNF Condition (For every $X \twoheadrightarrow Y$:

X MUST be Super Key):**

- In C

D : Determinant C is NOT a Super Key ($C^+ = \{A, C, D\}$ in R).

- In D

A : Determinant D is NOT a Super Key ($D^+ = \{A, D\}$ in R).

- **Relation FAILS BCNF!**

4. Conclusion: Relation is in 3NF but NOT in BCNF.

3. AKTU PYQ 3 (10 Marks — 2020-21 End-Sem)

Question:

Explain Lossless Join Decomposition and Dependency Preservation with suitable examples. State the necessary condition for a binary decomposition to be lossless.

Solution:

*(Complete formal definitions, the binary condition theorem $(R_1 \cap R_2)$

$\rightarrow R_1 \cup (R_1 \cap R_2)$

$\rightarrow R_2$, and tabular counter-example demonstrating spurious tuples as detailed in Module 09).*

4. Architectural Diagram
